DAFFODIL INTERNATIONAL UNIVERSITY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

FYDP (PHASE-I) EVALUATION REPORT

REPORTING PERIOD – SUMMER 2026

Project Identification:

I. Project Title	**Agreement-Stratified Evaluation of Deep Learning for Anatomical Landmark Recognition in Upper Gastrointestinal Endoscopy**	
II. Group Members	Name: Fatin Sadab Nibirh Student ID: 0242310005101526 Name: MD Himel Rahman Student ID: 0242310005101800	
III. Supervisor	Name: Ms. Shayla Sharmin Designation: Assistant Professor	
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V. Submission Date:	03-08-2026	
VI. Certificate:	"This is to certify that the final year design project work until Phase-I evaluation held on __________________, titled as stated in *Sec. I*, executed by the students' group mentioned in *Sec. II*, have been found satisfactory and every section of this report is reflecting the same."	*(Signature of Supervisor & date)*

Project Insights

Thematic Area(s): ***[Just click the check box]***	Artificial Intelligence and Machine Learning	☒
	Deep Learning	☒
	Health Informatics	☒
	Cybersecurity	☐
	Software Engineering and Development	☐
	Blockchain Technology	☐
	Internet of Things (IoT)	☐
	Computer Networks	☐
	Computer Vision	☒
	Natural Language Processing (NLP)	☐
	Robotics	☐
	Game Development	☐

	Cloud Computing	☐
	Image Processing	☒
	Others ***(please specify)*****:**	
Software packages, tools, and programming languages	Python 3.14.5; PyTorch 2.12.0+cu126 with CUDA 12.6; torchvision (ConvNeXt-Tiny, EfficientNet-B0, ImageNet-pretrained weights); NumPy; pandas; scikit-learn; SciPy; Pillow; Matplotlib; python-docx and python-pptx for automated report and slide generation; Git for version control; NCBI E-utilities for the literature search.	

CO Description for FYDP-Phase-I

CO	CO Descriptions	PO
CO4	Perform economic evaluation, cost estimation, and apply suitable project management procedures throughout the FYDP lifecycle in the context of developing the “Agreement-Stratified Evaluation of Deep Learning for Anatomical Landmark Recognition in Upper Gastrointestinal Endoscopy” project.	**PO11**
CO6	Select and apply appropriate methodologies, resources, and contemporary engineering/IT tools for prediction, modeling, and solving complex engineering processes for the “Agreement-Stratified Evaluation of Deep Learning for Anatomical Landmark Recognition in Upper Gastrointestinal Endoscopy” project.	**PO5**
CO7	Assess societal, health, safety, legal, and cultural issues and responsibilities in professional engineering practice related to the FYDP problem.	**PO6**
CO10	Operate effectively as an individual and as a member/leader in multidisciplinary teams during FYDP.	**PO9**

1. Project Overview

1.1 Introduction

Upper gastrointestinal endoscopy is the primary screening modality for gastric cancer, and its diagnostic yield depends on whether the operator inspects the whole mucosal surface rather than a convenient subset of it. The Systematic Screening protocol for the Stomach formalises that requirement as a fixed photographic sequence covering twenty-two anatomical landmarks, organised as a grid of four gastric walls crossed with six depth stations. A model that recognises the landmark visible in each frame can therefore act as a real-time coverage monitor and warn the operator that a region has not yet been visited.

Published systems report macro F1 near 85 to 88 on this recognition task, and the problem is generally treated as solved. Every one of those figures, however, is measured only on frames whose expert annotators all chose the same label. In the corpus used here that unanimous subset is 60.2 per cent of the data; the remaining 39.8 per cent, 3,516 images, is removed before scoring and never enters a reported result. The discarded fraction is not random noise. Half of all annotator conflicts place two experts on different walls of the same station, which means the disagreement follows the anatomy of the label space rather than scattering across it.

The main goal of this project is therefore to measure what current accuracy figures actually describe. It evaluates landmark classifiers across strata of expert agreement rather than on unanimous frames alone, and it tests whether training targets built from the complete four-annotator vote distribution improve accuracy, calibration and uncertainty behaviour on contested images. The contribution is diagnostic rather than architectural: no new network is proposed, and the improvement sought is in what is known about the task rather than in a leaderboard position.

Three specific objectives follow from that goal. The first is to establish that the corpus is sound and the training pipeline is correct, by auditing provenance and integrity before any model is fitted and then reproducing the published baseline to within a tolerance fixed in advance. The second is to quantify how performance and confidence behave as expert agreement falls, with intervals that respect the clustering of images within patients. The third is to build the study as a reproducible artefact, in which every figure, table and interval regenerates from committed scripts rather than being transcribed by hand.

This report covers the first four of the seven phases into which the work is divided. At the time of writing, the corpus of 8,834 images from 387 patients has passed an eight-criterion integrity gate, a systematic literature review has been completed, a baseline model has been trained, validated and tested, and its behaviour has been measured across four agreement strata. Figure {fig:workflow} shows the phase structure and marks the portion reported here.

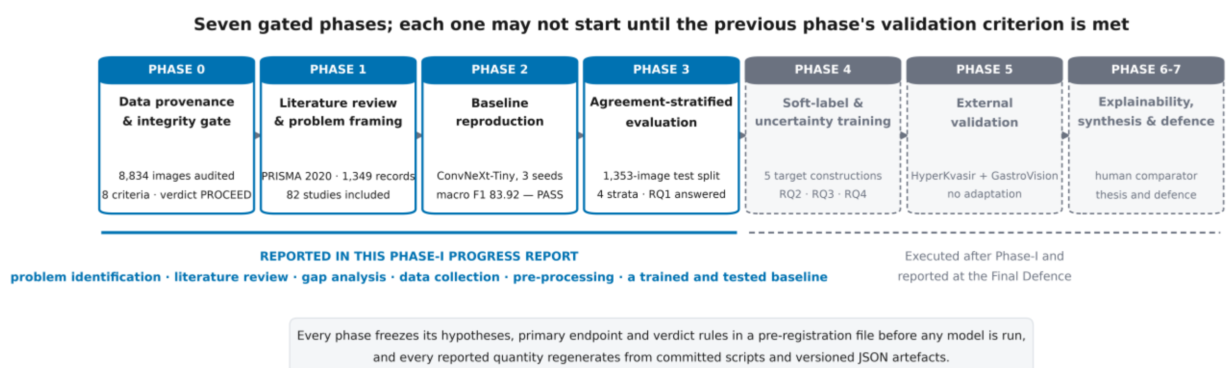

Figure 1. *Phase structure of the project. Each phase is gated on the previous phase's validation criterion, and the four phases reported here cover every Phase-I requirement.*

1.2 Background Study

A systematic review was carried out to establish what is already known and where the evidence stops. Seven themed queries were run against PubMed/MEDLINE, covering landmark recognition, endoscopic quality and blind-spot auditing, inter-observer variability, noisy and multi-annotator labels, uncertainty and calibration, external validation, and reporting standards for medical artificial intelligence. The searches returned 1,382 records, 1,349 after de-duplication, of which 82 studies were included: 68 from the database search and 14 added through the PRISMA other-methods arm, because several relevant computer-science venues are not indexed in MEDLINE. The flow is reported in Figure {fig:prisma} according to PRISMA 2020 [14].

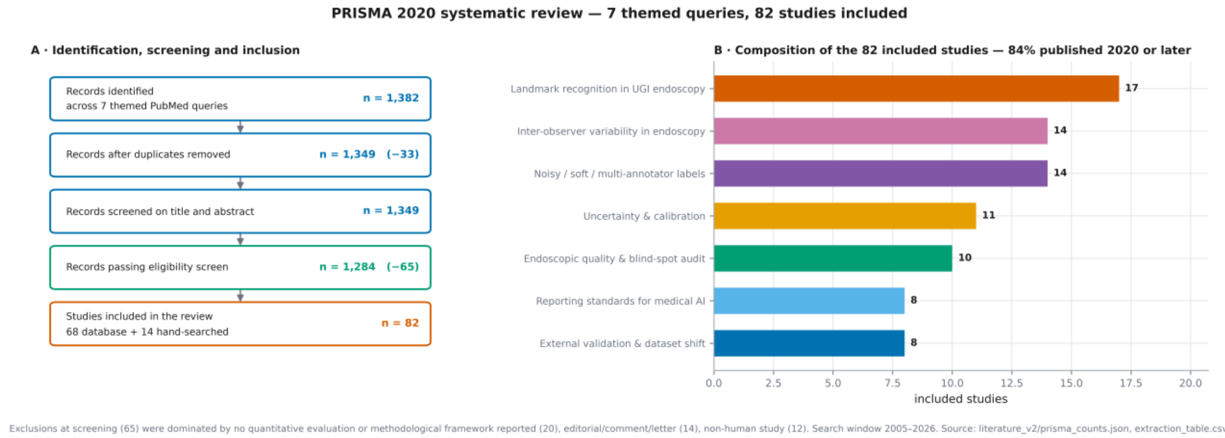

Figure 2. *PRISMA 2020 flow and the thematic composition of the included studies.*

Clinical task. The Systematic Screening protocol prescribes a fixed photographic route through the stomach. Its label space is the product of two axes: the circumferential wall, comprising the greater curvature, the anterior wall, the lesser curvature and the posterior wall; and the axial station, running from the antrum through to the final retroflex view. Automated recognition of this route underpins blind-spot monitoring and endoscopic quality scoring, and comparable systems have been reported to reduce blind-spot rates in prospective use [10].

Corpus and published baselines. GastroHUN [1] releases 8,834 images from 387 patients recorded at the Hospital Universitario Nacional de Colombia. Each image carries 4 independent labels over 23 classes, supplied by two gastroenterologists and two fellows, and the official splits are patient-disjoint. Table {tbl:baselines} lists the baselines the descriptor reports. Two features of that table matter more than the headline numbers. Every figure in it is measured on the unanimous subset alone. And changing only how the four annotator labels are combined moves macro F1 by 2.2 points, which is a larger effect than the gap between the smallest and largest architecture families the descriptor evaluates. The descriptor records that observation and does not pursue it; it is the direct precedent for this work.

Table 1. *Published baselines on this corpus, as reported by the dataset descriptor [1]. Every figure is measured on the complete-agreement subset only.*

Model or label construction	Parameters	Macro F1
ConvNeXt-Large	200 M	88.25 ± 0.22
ConvNeXt-Tiny	28 M	≈ 85
ResNet-152	60 M	85.28 ± 0.27
ConvNeXt-Tiny, fellow-agreement labels	28 M	87.05 ± 0.21

Model or label construction	Parameters	Macro F1
Best single annotator (G1) as target	—	84.82 ± 0.23
Human expert band	—	77.47 – 84.82

Agreement structure measured during preparation. Chance-corrected agreement over the released labels is Fleiss' kappa = 0.7475. Krippendorff's alpha and Gwet's AC1 coincide with it to four decimal places, because the screening protocol keeps the class marginal near-uniform and the three chance corrections become algebraically equal in that regime; the well-known kappa paradox therefore does not arise on this corpus. Seniority does not predict agreement: each fellow agrees more closely with either gastroenterologist than with the other fellow, so any design that treats a team as a coherent unit is unsupported by the data. Unanimity is 60.2 per cent, and 614 images (6.95 per cent) admit no majority label under any voting rule. Figure {fig:agreement} reports the cascade, the vote patterns and the pairwise agreement.

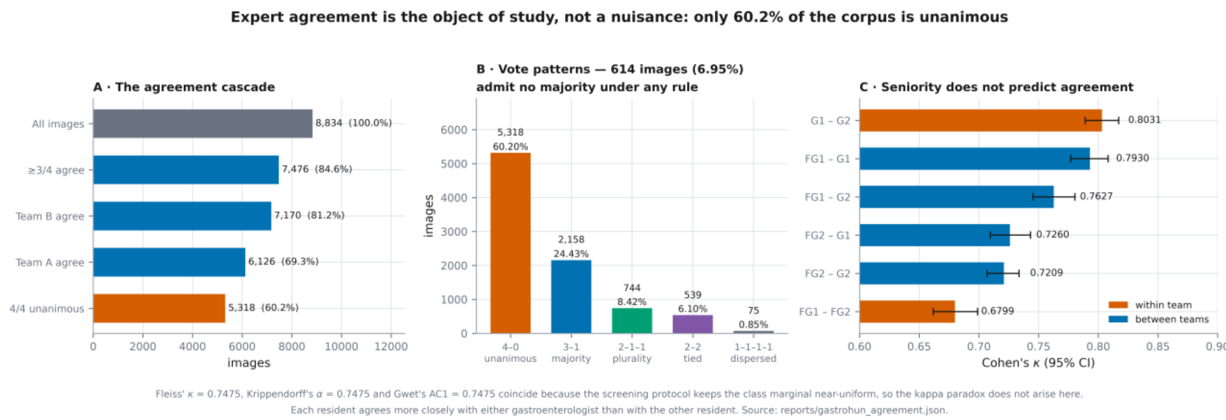

Figure 3. *Annotator agreement: the cascade from all images to unanimity, the distribution of vote patterns, and pairwise Cohen's kappa with patient-clustered intervals.*

Disagreement respects the anatomy. Of 12,800 pairwise conflicts, 50.96 per cent are same-station different-wall and only 19.91 per cent are same-wall different-station. Collapsing the label space to stations alone raises mean pairwise kappa from 0.7476 to 0.8597, whereas collapsing to walls alone changes it only to 0.7481. Endoscopists agree on how deep the scope is and disagree about which way it points. Figures {fig:labelspace} and {fig:disagreement} set out the label space and this decomposition. The structure matters for the design of the work, because a disagreement that follows a known geometry is something a model can be asked to reproduce, whereas unstructured noise is not.

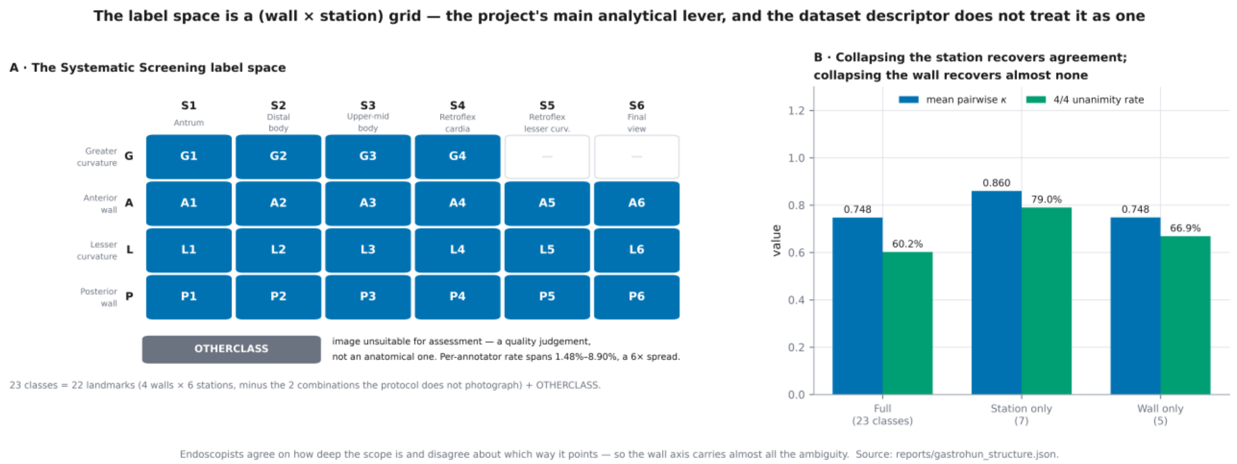

Figure 4. *The wall-by-station label space, and what happens to agreement when each axis is collapsed in turn.*

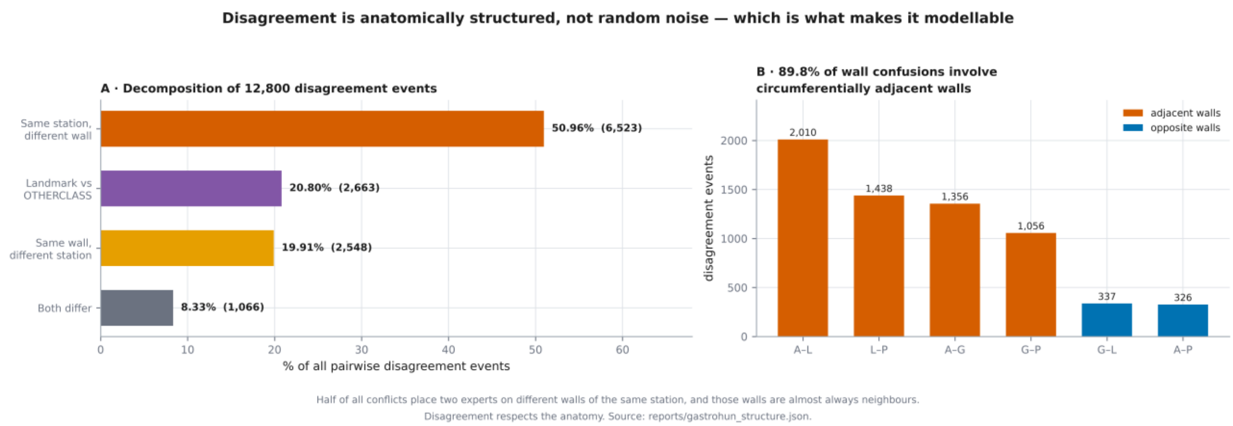

Figure 5. *Decomposition of the pairwise disagreement events, and the wall pairs that account for them.*

Inter-observer variability in endoscopy. Substantial rater disagreement is documented across endoscopic and histological classification, including the Paris classification of colorectal lesions [3], OLGA and OLGIM gastritis staging [4], and oesophageal varices in children [5]. Agreement is thus a property of the task rather than a defect of a single dataset, which is why an evaluation conditioned on unanimity reports a systematically easier problem than the clinical one.

Learning from multiple annotators. Modelling label confidence rather than a single consensus target has been shown to improve histology classification [6], and annotator uncertainty used as privileged information improves clinical prediction [7]. Multi-rater calibration is an active benchmark target [8]. These methods are rarely applied to endoscopic landmark recognition, largely because per-annotator labels are seldom released, which is precisely what makes this corpus suitable.

Reporting quality. Systematic reviews of diagnostic deep learning find that calibration and external validation are routinely omitted and that comparison against clinicians is weakly reported [2, 9]. Where external validation is performed, sizeable accuracy loss across centres is the norm, and modern networks are known to be poorly calibrated even when accurate [15]. This project therefore treats calibration and a change of centre as primary endpoints rather than as optional extras, and follows the CLAIM, TRIPOD+AI, STARD-AI, PROBAST+AI and PRISMA 2020 reporting standards.

1.3 Gap Analysis

Three gaps emerge from the background study. They are stated here as measurable deficiencies rather than as general observations, so that each one has a corresponding endpoint later in the design.

Gap 1: evaluation is conditioned on expert unanimity. No published evaluation on this corpus reports performance as a function of expert agreement, because contested images are filtered out before scoring. The reported figure therefore describes 60.2 per cent of the data and is silent about the rest. A deployed system meets the contested 39.8 per cent continuously and receives no signal that its validation excluded those frames. Its operating accuracy is consequently unknown on exactly the images where an automated second reader would be most useful.

Gap 2: annotator disagreement is discarded rather than used. Because per-annotator labels are rarely released, the standard pipeline collapses the vote distribution to a single label at ingest. On this corpus that discards a structured signal: 50.96 per cent of the 12,800 conflicts are same-station different-wall, and the walls involved are almost always circumferential neighbours. The descriptor's own finding that the label-combination rule moves macro F1 by more than the architecture family does confirms that the discarded information is worth something, but no controlled test of it has been reported.

Gap 3: calibration is reported rarely, and never by stratum. Reviews of diagnostic deep learning identify absent calibration reporting as one of the commonest omissions in this literature. Nothing in the published work establishes whether confidence estimated on unanimous frames remains trustworthy on ambiguous ones. This is the most consequential omission clinically: a model that stays confident where four experts could not agree produces assured errors, which is the failure mode least likely to be caught by an endoscopist working under time pressure.

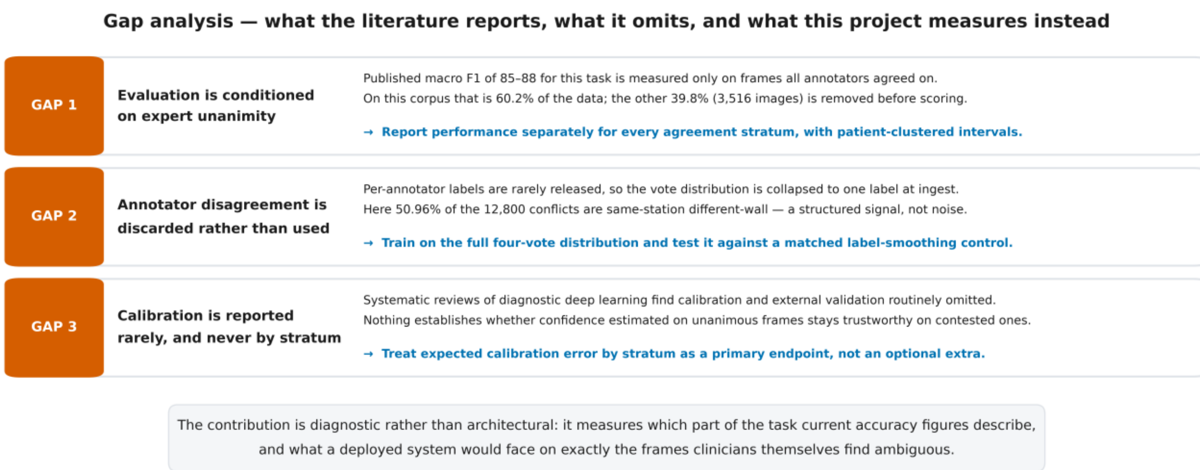

Figure 6. *Gap analysis: the three deficiencies identified in the background study, the evidence for each, and the design response adopted.*

A fourth, methodological gap should be recorded because it shapes how results are reported. Intervals in this literature are commonly computed by resampling images. Images from one patient are not independent observations here: per-patient agreement measured during preparation has a mean of 0.7459 with a standard deviation of 0.1448 and a range from 0.0694 to 1.0000. Every interval in this project therefore resamples patients rather than images, and is correspondingly wider than intervals published on the same quantity.

1.4Objectives

The objectives below are the ones this phase set out to achieve. Each is stated so that it can be judged met or unmet against evidence, and each has a corresponding artefact in the project repository.

1. Audit corpus provenance and integrity before any modelling: verify licence, ethics approval and documented consent; confirm that every image decodes and that the manifest and the file system agree; and establish, by a calibrated perceptual scan rather than an assumed threshold, that no image recurs across the patient-disjoint splits.

2. Complete a systematic literature review to PRISMA 2020 across the seven themes that bound the problem, and derive from it a gap statement naming measurable deficiencies rather than general observations.

3. Construct the pre-processing chain — annotation handling, cohort selection, resampling, normalisation, augmentation and the transfer-learned representation — and fix it once, so that every later phase differs from the baseline only in the quantity under test.

4. Train, validate and test at least one baseline model, and reproduce the published result on the unanimous subset to within a tolerance of ±1.5 macro F1 points fixed before training, so that later findings are attributable to the design rather than to a faulty pipeline.

5. Quantify macro F1, expected accuracy and expected calibration error separately for each agreement stratum of the official test split, with patient-clustered bootstrap intervals, and compare the resulting decline against the between-architecture difference the descriptor reports.

6. Release a reproducible pipeline in which every table, figure and interval in this report regenerates from committed scripts and versioned artefacts, with no value transcribed by hand.

2. Methodology/ Requirement Specification:

2.1 Research Design/ Prototype Design

The study is an evaluation-and-training design executed in seven gated phases, of which four are reported here. A phase may not begin until the preceding phase has met a validation criterion written down before it ran. Each phase fixes its hypotheses, its primary endpoint and its verdict rules in a pre-registration file that the generating script refuses to overwrite. This matters because most endpoints in this project are comparisons that could be made to favour either conclusion by a choice of scale or stratum taken after seeing the data; fixing the choice in advance removes that freedom.

The backbone is ConvNeXt-Tiny [13], initialised from ImageNet-1k weights and fitted with a 23-way linear head. It was selected on two grounds. It is the architecture whose published result on this corpus the project must reproduce, which is what makes the reproduction check meaningful; and at 27.8 million feature parameters it fits the 4 GB of video memory available on the project hardware, whereas the 200-million-parameter ConvNeXt-Large that tops the published table does not. ConvNeXt-Large is therefore cited as the published ceiling and deliberately not attempted.

Training follows a two-stage transfer schedule. The backbone is first frozen entirely and only the head is fitted, for 10 epochs at a constant learning rate of 0.001. The top 3 of the 8 feature modules, which hold 94.5 per cent of the feature parameters, are then unfrozen and fine-tuned at 0.0001 under cosine decay, with early stopping on validation macro F1 and a patience of 10 epochs. Model selection uses validation macro F1 and nothing else; the test split is touched once, after selection. Figure {fig:arch} shows the architecture and the schedule together.

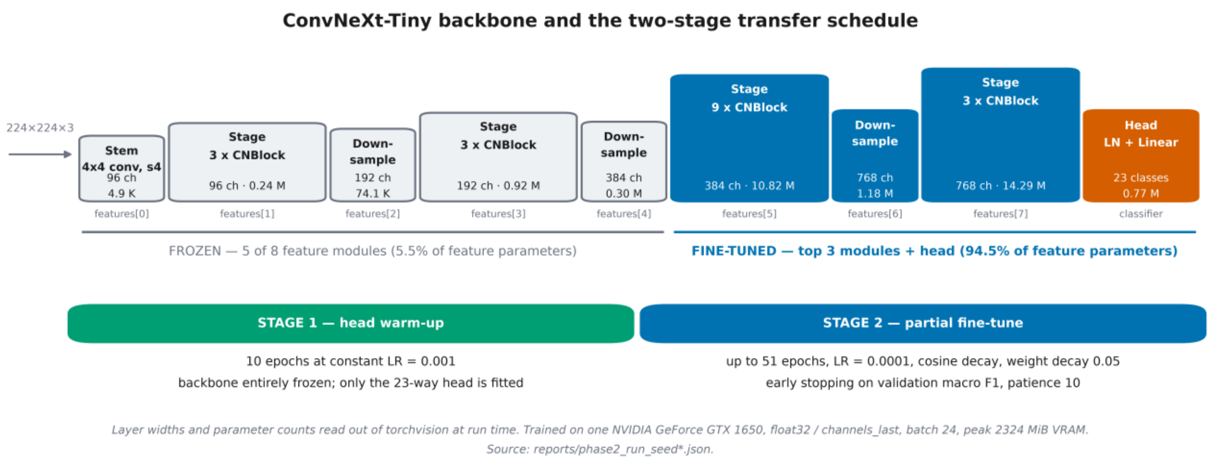

Figure 7. *The ConvNeXt-Tiny backbone, the parameter blocks held frozen, and the two-stage transfer schedule. Layer widths and parameter counts are read from the framework at run time.*

Table 2 records the configuration in full. One entry departs from the original plan and is declared rather than passed over. The plan specified mixed-precision float16 training with gradient scaling. Measured on this device, float16 ran at 0.38 times the throughput of float32, because the GPU is a part without tensor cores; float32 with a channels-last memory layout was therefore adopted. Numerical precision is a throughput decision rather than a scientific one, and float32 is the safer of the two, so the deviation carries no consequence for the result. It is recorded in the pre-registration as deviation DEV-1.

Table 2. *Model and training configuration. Every value is read from the committed run records rather than from the plan.*

Item	Setting
Backbone	ConvNeXt-Tiny, ImageNet-1k (IMAGENET1K_V1)
Classification head	LayerNorm + Linear, 23 classes
Input	224 × 224 × 3 RGB
Stage 1	10 epochs, head only, LR 0.001 constant
Stage 2	up to 51 epochs, top 3/8 feature modules, LR 0.0001 cosine decay
Weight decay	0.05
Early stopping	validation macro F1, patience 10
Batch size	24 (effective 24)
Precision	float32, channels_last (deviation DEV-1)
Seeds	1, 2, 3 — every result is a three-seed mean
Hardware	NVIDIA GeForce GTX 1650, 4 GB; peak 2324 MiB used
Software	Python 3.14.5, PyTorch 2.12.0+cu126, CUDA 12.6
Total training time	150 minutes for all three seeds

The cohort for this phase is the unanimous subset: 3,722 training, 793 validation and 803 test images, drawn from the official patient-level splits so that no patient appears in more than one split. That restriction is deliberate and temporary. It reproduces exactly the condition under which the published baseline was measured, which is what makes the reproduction check interpretable; the contested images are retained and brought back for the stratified evaluation reported in Section 3.2.

2.2 Data Collection/ Need Assessment

No new data was collected and no participant was recruited. The requirement this project has of a corpus is unusual, and it excludes almost every public alternative: the labels of the individual annotators must be released separately, because the object of study is the disagreement between them. A corpus that publishes only a consensus label cannot support any endpoint in this design.

GastroHUN [1] meets that requirement. It was obtained from its Figshare release under a CC BY 4.0 licence; ethics approval CEI-2019-06-10 and documented informed consent are recorded in the peer-reviewed descriptor. A licence discrepancy between the descriptor text and the Figshare record was identified during the audit and resolved in favour of the more permissive Figshare statement. The payload is 2.71 GB comprising 8,834 JPEG images across 387 patient directories. Table {tbl:corpus} profiles it.

Table 3. *Corpus profile, measured from the release rather than quoted from the descriptor.*

Property	Value
Images	8,834 (all JPEG, all RGB)
Patients	387
Classes	23 — 22 landmarks plus OTHERCLASS
Annotators	4 independent, retained separately (FG1, FG2, G1, G2)
Images per patient	22.83 ± 2.88 (range 7–32)
Resolutions	1350x1080 (8,427 images) · 900x720 (407 images)
Acquisition streams	direct capture 8,053 · video frame 781
Payload	2.71 GB
Official splits	patient-level; 269 train / 58 validation / 58 test patients
Licence and ethics	CC BY 4.0; ethics approval CEI-2019-06-10; documented informed consent

The corpus was then put through an eight-criterion integrity gate before any model was fitted, on the principle that a dataset should be shown to be sound rather than assumed to be. All 8,834 images decoded; there were 0 files missing from disk, 0 orphan files, 0 corrupt files and 0 folder-to-manifest mismatches. The gate returned PROCEED, with six criteria passing without qualification and two returning CONDITIONAL. A conditional verdict identifies a constraint that must be carried forward as a declared limitation rather than a defect that blocks the work: in this case, that 22 of 23 classes are too small to support per-class claims, and that the release contains no age or sex field, so no demographic subgroup or fairness claim is possible. Figure {fig:gate} summarises the gate.

Phase 0 data-integrity gate — verdict PROCEED (6 PASS, 2 CONDITIONAL)

G1	Provenance, ethics, licence	PASS	peer-reviewed descriptor; ethics CEI-2019-06-10; consent; CC BY 4.0
G2	Physical integrity	PASS	8,834/8,834 decoded · 0 missing · 0 orphan · 0 corrupt
G3	Duplication & contamination	PASS	0 exact duplicates; 39,015,361-pair perceptual scan → 0 cross-split duplicates
G4	Label architecture	PASS	4 annotators retained separately across 23 classes
G5	Agreement quantified	PASS	Fleiss κ = 0.7475; all 6 pairwise κ with patient-clustered CIs
G6	Split integrity	PASS	0 patient overlaps; class χ^2 p = 0.999997; per-patient κ Kruskal–Wallis p = 0.982
G7	Statistical power	CONDITIONAL	22/23 classes exceed a ±10 pp Wilson half-width → per-class claims are exploratory (L1)
G8	Population description	CONDITIONAL	no age or sex anywhere in the release → no demographic or fairness claim is possible (L2)

An uncalibrated threshold is not a measurement. The contamination scan first reported 54 cross-split duplicate pairs; anchoring the decision rule on a synthetic-duplicate positive control (4,000 pairs) and a class-matched null (4,000 pairs) reduced that to 0, and visual audit confirmed the flagged pairs were different patients photographed at the same landmark.

Figure 8. *The eight-criterion data-integrity gate and its verdicts.*

The contamination scan deserves a separate note, because it produced the most instructive result of the audit. An exhaustive perceptual comparison of all 39,015,361 image pairs, followed by pixel verification, initially reported 54 cross-split duplicate pairs under a threshold chosen by convention. That number was an artefact of the threshold. A first attempt to calibrate it against randomly paired images was also wrong, because random pairs mostly compare different anatomical stations and therefore set an artificially low bar. Rebuilding the null as class-matched pairs — two different patients photographed at the same landmark, 4,000 of them — and anchoring the decision rule on a synthetic-duplicate positive control of 4,000 pairs reduced the count to 0. Visual audit confirmed that the flagged pairs were different patients photographed at the same landmark. An uncalibrated threshold is not a measurement, and a threshold calibrated against the wrong comparison is not much better; this standard is applied to every threshold used later in the project.

Split integrity was verified independently of the descriptor's claims. There are 0 patient overlaps between any two official splits. Class composition does not differ across splits (chi-square p = 0.999997), and neither does per-patient agreement (Kruskal–Wallis p = 0.982), so the test split is not systematically easier or harder than the training split. One imbalance was found and is declared: the acquisition stream is unevenly distributed across splits (chi-square p = 1.9e-22), so per-split composition is reported and a sensitivity analysis excluding video-derived frames is run wherever the stream could plausibly matter.

2.3 Analysis Techniques

Pre-processing is fixed once, in this phase, and reused unchanged thereafter. Figure 9 shows the whole chain. Three of its stages carry decisions worth stating explicitly, because the obvious choice is wrong in each case.

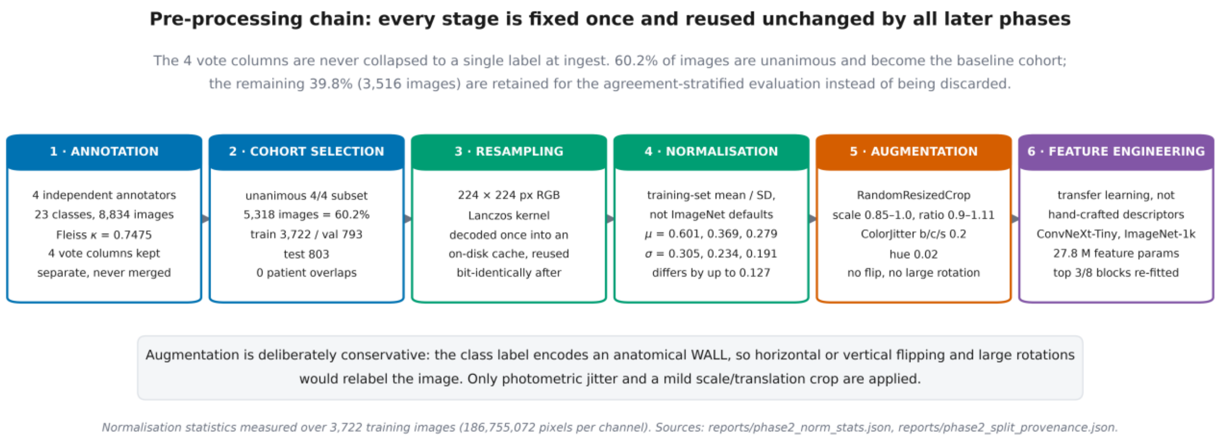

Figure 9. *The pre-processing chain, from annotation through to the transfer-learned representation.*

Annotation. The 4 vote columns are never collapsed to a single label at ingest. Annotation is treated as a pre-processing stage that produces four parallel label channels, and each downstream phase decides for itself how to combine them. This is what makes the agreement-stratified evaluation possible at all, because the conventional pipeline destroys the required information in its first step. The unanimous subset becomes the baseline cohort; the contested images are retained rather than discarded, and are tagged with the agreement tier derived from the vote matrix.

Resampling and normalisation. Images are resampled once to 224 × 224 pixels with a Lanczos kernel and written to a cache that every later phase reads, so that no phase can silently introduce a different decode path; the cache is verified bit-identical whenever it is extended. Normalisation uses statistics measured on this training set — a mean of (0.601, 0.369, 0.279) and a standard deviation of (0.305, 0.234, 0.191), over 3,722 images and 186,755,072 pixels per channel — rather than the ImageNet defaults, which differ by as much as 0.127 in the red channel. Endoscopic imagery is dominated by mucosal red under a point light source and does not resemble the natural-image distribution those defaults were computed on.

Data augmentation. The augmentation policy is deliberately conservative, and the reason is anatomical rather than statistical. The class label encodes a gastric wall, so a horizontal or vertical flip, or a large rotation, changes the apparent wall and therefore silently relabels the image. The standard augmentation recipe for natural images would inject label noise into precisely the axis that carries most of the annotator disagreement. Only a mild scale-and-translation crop (RandomResizedCrop, scale 0.85–1.0, aspect ratio 0.9–1.111) and photometric jitter (brightness, contrast and saturation 0.2, hue 0.02) are applied. Augmentation is applied to the training split only; validation and test images pass through normalisation alone. Table {tbl:preprocspec} records the policy.

Table 4. *Pre-processing and augmentation specification.*

Stage	Setting	Applied to
Decode	JPEG to RGB, verified against the SHA-256 inventory	all

Stage	Setting	Applied to
Resample	224 × 224, Lanczos kernel, cached once and reused	all
Normalise	training-set mean 0.601, 0.369, 0.279; SD 0.305, 0.234, 0.191	all
Geometric augmentation	RandomResizedCrop, scale 0.85–1.0, ratio 0.9–1.111	train only
Photometric augmentation	ColorJitter, brightness/contrast/saturation 0.2, hue 0.02	train only
Excluded by design	horizontal flip, vertical flip, rotation beyond the crop — each changes the apparent gastric wall and would relabel the image	—

Feature engineering. No hand-crafted descriptors are computed. The feature representation is learned by transfer from ImageNet-1k and then partially re-fitted to this domain: the lower 5 feature modules are held frozen as a generic low-level feature extractor, and the top 3 modules, holding 94.5 per cent of the feature parameters, are adapted to the endoscopic domain. Freezing the whole backbone leaves the representation unable to model mucosal texture; unfreezing all of it overfits a corpus of this size. The split point was specified before training as the top 40 per cent of feature modules and was not tuned afterwards.

Metrics and statistics. Macro F1 is the primary metric, matching the published baseline on a near-balanced 23-class problem. Secondary metrics are macro and weighted precision and recall, and accuracy. Calibration is reported as expected calibration error, maximum calibration error and the Brier score, with reliability diagrams. On strata where no single ground-truth label exists, two distribution-aware quantities replace accuracy: expected accuracy, the probability mass the model's single prediction captures under the vote distribution; and the any-annotator hit rate, an indicator of whether the prediction was named by at least one expert. Every interval reported anywhere in this project is a patient-clustered bootstrap over 1,000 resamples with a fixed seed, resampling the 58 test patients rather than the images. Per-class results are treated as exploratory throughout, for the power reason recorded at the integrity gate.

3. Progress Achieved:

3.1 Completed Tasks

The tasks below were completed during the reporting period. Each produced a committed artefact and each was gated on a criterion recorded before it ran. The gate results are given in the right-hand column of Table 5 rather than described in prose, so that a claim of completion can be checked rather than taken on trust.

Table 5. *Tasks completed during the reporting period, with the validation criterion each was gated on.*

#	Task	Validation result
1	Corpus viability assessment and replacement decision	12 defects catalogued in the previously selected corpus; replacement documented and approved

#	Task	Validation result
2	Corpus acquisition, provenance and licence verification	peer-reviewed descriptor, ethics CEI-2019-06-10, informed consent and CC BY 4.0 all confirmed
3	Physical integrity audit	8,834/8,834 images decoded; 0 missing, 0 orphan, 0 corrupt
4	Duplication and cross-split contamination scan	39,015,361 pairs examined; 0 cross-split duplicates after threshold calibration
5	Agreement quantification	Fleiss kappa = 0.7475; all six pairwise kappa with patient-clustered intervals
6	Split integrity verification	0 patient overlaps; class chi-square p = 0.999997
7	Systematic literature review to PRISMA 2020	1,349 unique records screened; 82 studies included across 7 themes
8	Pre-processing pipeline and image cache	224 px cache built and verified; normalisation statistics measured on 3,722 images
9	Pre-registration of the baseline experiment	frozen before training; target, tolerance, seeds, bootstrap procedure and diagnostic order all fixed in advance
10	Baseline model trained, validated and tested across three seeds	macro F1 83.92 against a published 85 ± 1.5 — verdict PASS
11	Agreement-stratified evaluation of the frozen checkpoints	2,409 predictions reproduced exactly across three seeds; four strata scored
12	Reproducible reporting pipeline	every figure and table in this report regenerates from committed scripts and JSON artefacts

3.2 Results Obtained

Detailed result analysis belongs to the next phase. What follows establishes that the pipeline is correct, and reports the one finding already firm enough to state.

Baseline reproduction. Three seeds were trained under identical conditions, differing only in initialisation and data order. All three stopped by early stopping rather than by the compute-imposed epoch cap, which means the schedule was not truncated. The three-seed mean macro F1 on the 803-image unanimous test set is 83.92 (95 per cent CI 81.47 to 86.20, patient-clustered bootstrap over 1,000 resamples), against a published 85 and a tolerance of ±1.5 points fixed before training. The difference is -1.08 points and the pre-registered verdict is PASS. Seed-to-seed spread is 0.92 points. Figure {fig:training} shows the training dynamics and the outcome against the acceptance band; Table {tbl:metrics} gives the full metric set.

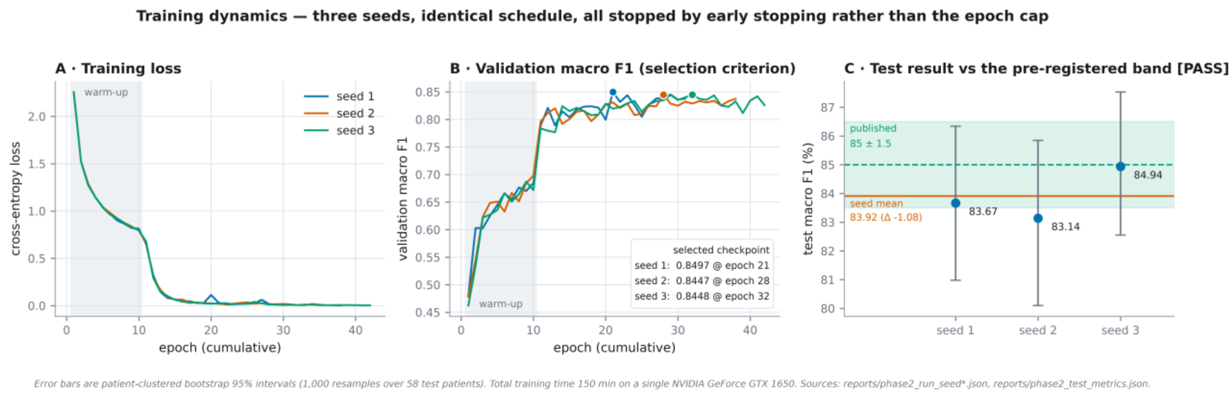

Figure 10. *Training dynamics for the three seeds, and the test result against the acceptance band fixed before training.*

Table 6. *Baseline performance on the unanimous test split, three-seed mean. Intervals are patient-clustered bootstrap 95 per cent intervals.*

Metric	Value
Macro F1	83.92 (95% CI 81.47–86.20)
Weighted F1	84.35
Macro precision	84.85
Macro recall	84.02
Accuracy	84.39
Expected calibration error	9.27 per cent
Brier score	0.2448
Seed spread (standard deviation)	0.92 points
Per-seed macro F1	83.67, 83.14, 84.94
Reproduction verdict	PASS — difference -1.08 against 85 ± 1.5

The reproduction being a PASS is the precondition for everything that follows, and it should be read narrowly. It says the pipeline — cohort construction, decode path, normalisation, schedule and evaluation — behaves as the published one did. It does not say the model is good, and the next result is that on most of the corpus it is not.

Agreement-stratified evaluation. The three frozen checkpoints were then evaluated, without retraining and without any threshold tuning, on the full 1,353-image official test split, stratified by how many of the four annotators agreed. As an internal consistency check, the predictions on the unanimous stratum reproduce the baseline predictions exactly — 2,409 comparisons across three seeds with 0 mismatches — which confirms the new evaluation path is wired correctly rather than producing a new result. Table {tbl:strata} and Figure {fig:strat} report the outcome.

Table 7. *Performance by agreement stratum, three-seed mean, from the frozen baseline checkpoints. Expected accuracy and the any-annotator hit rate are the distribution-aware metrics used where no single ground-truth label exists.*

Stratum	**Images**	**Patients**	**Macro F1**	**Expected accuracy**	**Any-annotator hit**	**Attainable ceiling**
4/4 unanimous	803	58	83.91	84.39	84.39	100.00
3/4 majority	342	57	48.95	50.71	80.02	74.23
2–1–1 plurality	127	38	26.15	27.82	72.70	44.55
2–2 and 1–1–1–1 pooled	81	39	30.79	38.07	79.83	40.23

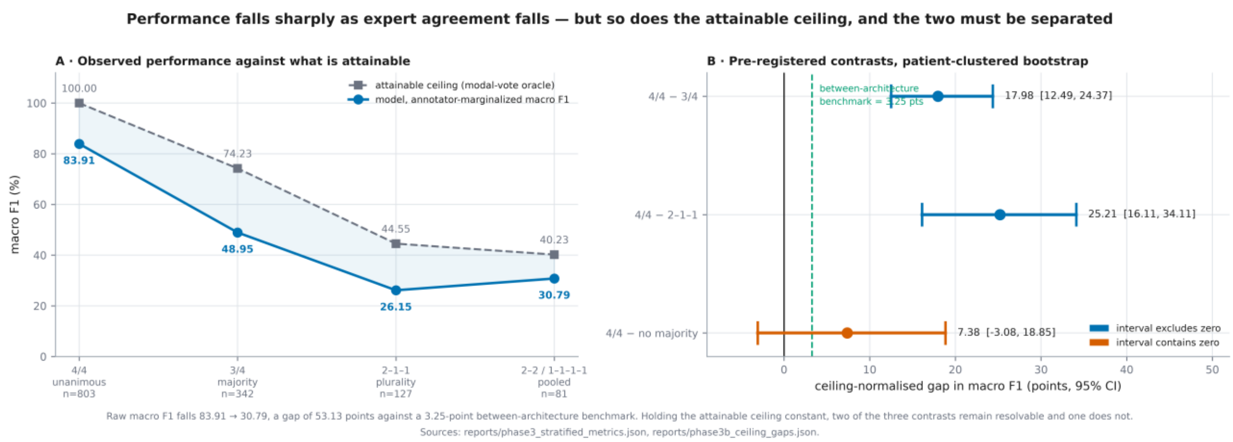

Figure 11. *Performance across agreement strata against the attainable ceiling, and the pre-registered contrasts with patient-clustered intervals.*

Macro F1 falls from 83.91 on unanimous images to 30.79 on images with no majority label, a gap of 53.13 points. For scale, the difference the descriptor reports between its smallest and largest architectures is 3.25 points. The choice of architecture is therefore a rounding error beside the choice of which images to evaluate on. The decline is not strictly monotonic — Spearman rho = -0.80, p = 0.20 over four tiers — and that non-monotonicity is reported rather than smoothed over.

A large part of that fall is not the model getting worse. On contested images the best achievable score is itself lower, because no single label can match a divided panel: the attainable ceiling falls from 100.00 to 40.23 across the same four strata. Holding that ceiling constant, the unanimous-to-majority gap is 17.98 points (95 per cent CI 12.49 to 24.37) and the unanimous-to-plurality gap is 25.21 points (95 per cent CI 16.11 to 34.11); both intervals exclude zero, so a real model shortfall remains after the ceiling is accounted for. The unanimous-to-no-majority contrast does not resolve, its interval containing zero, and is reported as unresolved. Separating a falling reference standard from a falling classifier, which the literature reports as one quantity, is the central analytical move of this project.

Calibration. The clearest and most consequential finding of the phase concerns confidence rather than accuracy. Expected calibration error rises from 9.15 per cent on unanimous images to 56.40 per cent on plurality images. The mechanism is visible in Figure {fig:calib}: mean predicted confidence falls only 9.34 points between those two strata while expected accuracy falls 56.57.

The model does not know it has entered harder territory. Clinically this is the failure mode that matters: a system which remains confident exactly where four experts could not agree offers no signal that its output should be doubted.

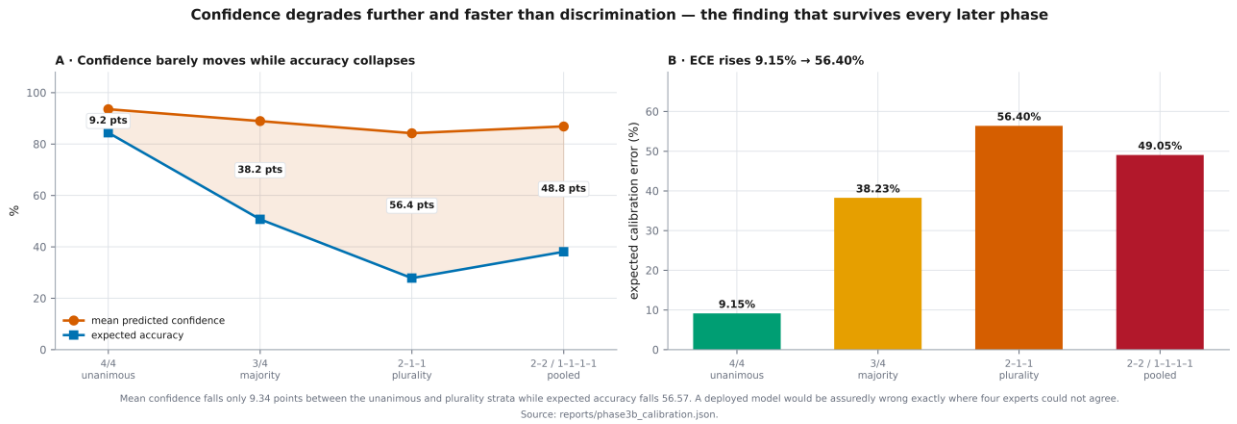

Figure 12. *Calibration by agreement stratum: predicted confidence against expected accuracy, and expected calibration error.*

Two confound checks were run against the stratified result and both were negative, which is what makes the effect attributable to agreement rather than to composition. Class composition explains between 3.5 and 4.2 per cent of the decline. Acquisition-stream composition does not differ across strata (chi-square $p = 0.298$), and restricting the analysis to the dominant stream shifts the curve by no more than 0.18 points. All twenty-three classes are populated in every stratum, so the macro average is not being distorted by absent classes.

4. Challenges Faced:

Four obstacles materially affected the work during this period. Each is recorded with the strategy adopted in Table 8; the two that changed the shape of the project are discussed first.

The first was that the originally selected corpus did not survive its own audit. A tabular dataset of endoscopy reports had been chosen for a natural-language design. Testing before modelling showed that the association between every feature and the target was indistinguishable from chance, the largest Cramér's V being 0.0614, and that the target had been constructed by pattern-matching over the same text fields that would have become the features. A classifier reached 100 per cent accuracy on it, and still 91.65 per cent after the most obvious leaking column was removed. By the data processing inequality, no amount of pre-processing can create signal that is not present. The project pivoted to imaging. The retired corpus was kept as a negative control against which the audit protocol itself can be tested, since an audit that passes everything it is shown is not an audit.

The second was hardware. The available GPU has 4 GB of video memory, which excludes the largest published architecture on this task outright. Rather than assume the recommended mixed-precision settings would help, throughput was measured on the actual device; float16 proved markedly slower than float32 on this part, so the plan was changed and the deviation recorded. The wider lesson — measure the machine rather than trust the guidance — was applied to every subsequent budgeting decision.

Table 8. *Issues and challenges encountered during the reporting period, with the strategy adopted for each.*

S. No.	Issues and challenges	Strategies or plans
1	The originally selected corpus had no learnable signal and circularly constructed labels, discovered during the pre-modelling audit.	Replaced the corpus with GastroHUN after a documented viability assessment, and retained the original as a negative control that validates the audit protocol itself.
2	Only 4 GB of GPU memory is available, and the recommended mixed-precision setting proved slower than full precision on this device.	Benchmarked the hardware instead of assuming; adopted float32 with a channels-last layout and recorded the departure as a declared deviation. ConvNeXt-Large is cited as the published ceiling and not attempted.
3	The cross-split contamination scan reported 54 duplicate pairs under a conventional threshold, which would have invalidated the official splits.	Rebuilt the decision rule against a class-matched null and a synthetic-duplicate positive control; the confirmed count fell to zero and was verified by visual audit.
4	No single-label ground truth exists on the tied and dispersed strata, so conventional accuracy metrics are undefined there.	Pre-specified two distribution-aware metrics, expected accuracy and the any-annotator hit rate, together with a pooling rule for the smallest strata, all fixed on corpus structure before any model output was seen.

5. Next Steps:

The tasks below carry the project from a measurement of the problem to an intervention on it. Each is scheduled against the phase structure in Figure 1 and each ends in a pre-registered verdict rather than an open-ended investigation.

Table 9. *Tasks and milestones planned for the next phase, with estimated completion dates.*

S. No.	Next task	Estimated completion (MM-YY)
1	Train and compare five target constructions: hard consensus labels, majority labels, vote-proportion soft targets, matched label smoothing as a regularisation control, and an anatomy-aware loss	09-26
2	Measure calibration and predictive uncertainty for every configuration, and correlate predictive entropy with annotator vote entropy within stratum	10-26
3	External validation on HyperKvasir and GastroVision without adaptation, with the label-space mapping frozen before any image is scored	11-26
4	Explainability and error analysis against a human comparator, and thesis preparation for the Final Defence	12-26

6. Updated Timeline:

Table 10 sets out the schedule against calendar weeks, with the estimated period above the actual period for each task. One adjustment to the original plan should be noted: the corpus replacement described in Section 4 consumed roughly two weeks that had been allocated to modelling, and that slippage is visible in the first two tasks. The time was recovered because the baseline reproduced at the first attempt, so the phase ends in the week originally planned.

Table 10. *Updated project timeline by calendar week. For each task the upper bar is the estimated period and the lower bar the actual period, keyed to the legend beneath the table.*

Tasks	Weeks																	
	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
Corpus audit, integrity gate and contamination scan																		
Literature review and gap analysis																		
Pre-processing pipeline, pre-registration and baseline training																		
Agreement-stratified evaluation and Phase-I reporting																		

Estimated Work Period	
Actual Work Period	

7. Resources Utilized:

Computation was carried out on a single workstation with an NVIDIA GeForce GTX 1650 holding 4 GB of video memory, alongside 16 GB of system memory. Peak memory use during training was 2324 MiB, and the three baseline runs took 150 minutes in total. No cloud or institutional compute was required, which was itself a design constraint: the whole study is reproducible on commodity hardware.

The software stack is Python 3.14.5 with PyTorch 2.12.0+cu126 on CUDA 12.6, torchvision for the pretrained backbones, and NumPy, pandas, scikit-learn and SciPy for evaluation and statistics. Figures are produced with Matplotlib and exported at publication resolution; this report and the defence slides are generated programmatically with python-docx and python-pptx from the same artefact set, which is why no number in either document is transcribed by hand. Literature retrieval used the NCBI E-utilities. Version control is Git, with each phase committed as a self-contained unit.

The data resource is the GastroHUN release: 2.71 GB of labelled images with per-annotator labels and official patient-level splits, obtained under CC BY 4.0. HyperKvasir [11] and GastroVision [12] have been acquired for the external validation planned in the next phase. The video release accompanying GastroHUN was deliberately not downloaded, as sequence modelling is out of scope.

8. Project Management and Financial Analysis:

The project is managed as a sequence of gated phases rather than as a continuous effort, and the gate is the unit of progress. Each phase begins by writing a plan that fixes its hypotheses, primary endpoint and verdict rules, and ends by producing a report together with the JSON artefacts from which that report regenerates. Work is divided between the two group members by phase component rather than by file, with the data pipeline and training code on one side and the evaluation, statistics and reporting layer on the other. Both members review every pre-registration before it is frozen, because a pre-registration reviewed after the fact is worthless.

The direct financial cost of the project to date is zero. The corpus is openly licensed, every software component is open source, and all computation runs on hardware already owned. The realistic cost is time: 150 minutes of GPU time for the baseline runs, and a substantially larger allocation for the next phase, where five configurations across three seeds each are planned. That workload was estimated by measuring the cost of one epoch on the actual device rather than by extrapolating from the baseline, because the anatomy-aware loss adds per-batch work a naive projection would miss. Table {tbl:cost} sets out the resource position.

Table 11. *Resource and cost position for the reporting period.*

Item	This period	Next period (estimated)
Direct monetary cost	BDT 0	BDT 0
Corpus licensing	CC BY 4.0, no fee	CC BY, no fee (two external corpora)
Software licensing	open source throughout	open source throughout
Compute	150 min GPU, 3 training runs	approximately 17 h GPU, 12 training runs
Hardware	NVIDIA GeForce GTX 1650, 4 GB, already owned	unchanged
Storage	2.71 GB corpus plus caches and checkpoints	additional external corpora and checkpoints
Principal risk	corpus replacement (materialised, absorbed)	training budget an order of magnitude larger on one GPU

The main scheduling risk carried forward is that the next phase's training budget is roughly an order of magnitude larger than this phase's, on the same single GPU. It is mitigated by running the configurations in seed-major order, so that a complete one-seed comparison across all five arms exists early and a truncated budget degrades the precision of the result rather than removing an arm from it.

9. Future Considerations:

Four considerations are likely to shape the next phase, and are recorded now so that they are not presented later as discoveries.

The contested strata are small. The plurality stratum holds 127 images from 38 patients, and the pooled no-majority stratum holds 81 images from 39 patients. Patient-clustered intervals on samples of that size are wide, and the unanimous-to-no-majority contrast already fails to resolve for this reason. The pooling rule was fixed in advance on corpus structure alone; if a contrast remains unresolved it will be reported as unresolved rather than re-cut until it separates.

A benefit from soft targets must be separated from ordinary regularisation. Training on the vote distribution softens the target, and softening a target is regularising whether or not the softening carries information about which classes are plausible. The comparison arm is therefore not the hard-label baseline but a label-smoothing control matched to the probability mass the soft target displaces from the modal label. Without that control any gain would be uninterpretable, and the control is not optional.

External validation may not be available in the form planned. The label space here is a product of wall and station; the external corpora identified for the next phase are not guaranteed to carry the wall axis at all. If they do not, a twenty-three-way external comparison is not available from them, and the endpoint will have to be reframed as a coarser anatomical collapse — declared before scoring, not after seeing that the fine-grained comparison disappoints.

Finally, the calibration finding needs a mechanism. Expected calibration error reaching 56.40 per cent on contested images could be a consequence of training only on unanimous frames, in which case training on the vote distribution should repair it; or it could be a property of the problem that survives every target construction. Those two possibilities have opposite implications for deployment, and distinguishing them is the single most useful thing the next phase can do.

10. Conclusion:

This phase set out to establish that the corpus is sound, that the pipeline is correct, and that the question the project asks is worth asking. All three are now supported by evidence rather than by argument. The corpus passed an eight-criterion integrity gate with a PROCEED verdict, the two conditional findings being declared as limitations rather than quietly absorbed. A systematic review of 1,349 unique records across 7 themes produced 82 included studies and a gap statement with three measurable components. The baseline model reproduced the published result at 83.92 macro F1 against 85 ± 1.5, a pre-registered PASS.

The substantive finding is that the reported accuracy of this task describes 60.2 per cent of it. Macro F1 falls from 83.91 to 30.79 across agreement strata, a 53.13-point spread beside a 3.25-

point difference between architecture families. Part of that fall is the reference standard weakening rather than the classifier failing, and the ceiling-normalised analysis separates the two and sizes each: a genuine model shortfall survives on the majority and plurality strata, while the pooled no-majority contrast does not resolve. Confidence degrades further and faster than discrimination, with expected calibration error rising from 9.15 to 56.40 per cent while mean confidence barely moves.

Every Phase-I requirement has been met and evidenced: problem identification, literature review and gap analysis in Section 1; data collection, annotation handling, augmentation and feature engineering in Section 2; and a baseline model trained, validated and tested in Section 3. The next phase turns from measuring the problem to acting on it, by testing whether the annotator disagreement the field discards can be used as a training signal, against a control designed so that a positive result cannot be confused with ordinary regularisation.

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Appendix

A. Reproducibility index. Every quantity in this report is generated by a committed script and stored in a versioned artefact before it is read into the document; no value is transcribed by hand. Table 12 maps the report's principal claims to the artefacts that carry them.

Table 12. *Reproducibility index: report claim mapped to generating script and stored artefact.*

Claim in this report	Generating script	Artefact
Corpus profile and physical integrity	src/data/gastrohun_inventory.py	reports/gastrohun_inventory.json
Agreement statistics and vote patterns	src/data/gastrohun_agreement.py	reports/gastrohun_agreement.json
Label-space structure and disagreement decomposition	src/data/gastrohun_structure.py	reports/gastrohun_structure.json
Contamination scan and threshold calibration	src/data/gastrohun_neardup.py, gastrohun_dup_calibration.py	reports/gastrohun_neardup.json, gastrohun_dup_calibration.json
PRISMA counts and included studies	src/literature/search_v2.py, eligibility_v2.py, enrich_v2.py	literature_v2/prisma_counts.json, extraction_table.csv
Normalisation statistics	src/models/phase2_normstats.py	reports/phase2_norm_stats.json
Cohort construction and split integrity	src/models/phase2_data.py	reports/phase2_split_provenance.json
Pre-registration of the baseline experiment	src/models/phase2_prereg.py	reports/phase2_prereg.json

Claim in this report	Generating script	Artefact
Training runs and realised schedules	src/models/phase2_train.py	reports/phase2_run_seed{1,2,3}.json
Baseline test metrics and reproduction verdict	src/models/phase2_eval.py	reports/phase2_test_metrics.json
Agreement-stratified metrics	src/models/phase3_eval.py	reports/phase3_stratified_metrics.json
Calibration by stratum	src/models/phase3b_calibration.py	reports/phase3b_calibration.json
Attainable ceiling and normalised gaps	src/models/phase3b_ceiling.py	reports/phase3b_ceiling_gaps.json
Every figure in this report	src/report/figures_phase1.py	figures_phase1/PH1_F01–F12 at 600 dpi
This document	src/report/build_phase1_docx.py with content_phase1_report.py and phase1_facts.py	Phase-I_Progress_Report.docx

B. Statistical procedure. Unless stated otherwise, every interval is a patient-clustered bootstrap 95 per cent interval computed over 1,000 resamples with seed 20260726, drawing 58 test patients with replacement and recomputing the statistic on each resample. Images are never resampled directly, because per-patient agreement varies systematically and images within a patient are therefore not independent observations. Results reported as a three-seed mean are the mean of the per-seed point estimates, with the interval computed on the seed-mean statistic.

C. Declared limitations carried into the next phase. Twenty-two of the twenty-three classes are underpowered for per-class claims, so the primary metric is macro-averaged and per-class results are exploratory. The release records no age or sex, so no demographic subgroup or fairness claim is possible. The corpus is single-centre and single-vendor, which makes external validation necessary rather than optional. The acquisition stream is imbalanced across splits, so per-split composition is reported and a sensitivity analysis excluding video-derived frames is run. Clinical context exists for only 60.2 per cent of patients, so no claim links landmark performance to disease status. Four annotators is a small panel, so vote proportions are treated as an ordinal ambiguity signal rather than as a calibrated probability.

FINAL YEAR DESIGN PROJECT
PHASE-I PROGRESS REPORT

This report, in the form of a template, has been specifically designed for BSc. students working on their Final Year Design Project (FYDP) at Computer Science and Engineering Department, Daffodil International University (DIU).

Every group of students is required to do the following:

1. Complete all the sections of this template
2. Get it certified by the assigned supervisor before one week of Phase-I evaluation presentations
3. Submit 01 photocopy to each of the following, on or before the day of Phase-I presentations:
    a. Supervisor
    b. Internal Evaluator
4. Submit original copy to FYDP committee on the day of Phase-I presentations.

Note:

1. Use English
2. There should be <u>NO</u> grammatical or spelling mistakes
3. Submission after due date will not be accepted
4. For more information, contact your Supervisor

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